

SIMATS ENGINEERING

CO4_AT1- Code Challenge

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA0613

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Q11. Longest Common Subsequence (LCS) — C Code

1. Source Code

```
#include <stdio.h>

#include <string.h>


// Function to find LCS length
int lcs(char s1[], char s2[]) {
    int m = strlen(s1);
    int n = strlen(s2);


    // DP table
    int dp[m + 1][n + 1];


    // Initialize first row and first column
    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0)
```

```

        dp[i][j] = 0;
    }
}

// Fill DP table
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {

        if (s1[i - 1] == s2[j - 1]) {
            dp[i][j] = dp[i - 1][j - 1] + 1;
        }
        else {
            if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j];
            else
                dp[i][j] = dp[i][j - 1];
        }
    }
}

return dp[m][n];
}

int main() {
    char s1[100], s2[100];

    printf("Enter first string: ");

```

```
scanf("%s", s1);

printf("Enter second string: ");
scanf("%s", s2);

int result = lcs(s1, s2);

printf("LCS Length = %d\n", result);

return 0;
}
```

Sample Input

Enter first string: ABCDGH

Enter second string: AEDFHR

Sample Output

LCS Length = 3

One possible LCS is ADH.